6.4 D'Alembert's Principle in curvilinear translation

The equations of curvilinear translation can be written as

$$\Sigma F_x + \left(-\frac{W}{g}\ddot{x}\right) = 0 \quad and \quad \Sigma F_y + \left(-\frac{W}{g}\ddot{y}\right) = 0$$

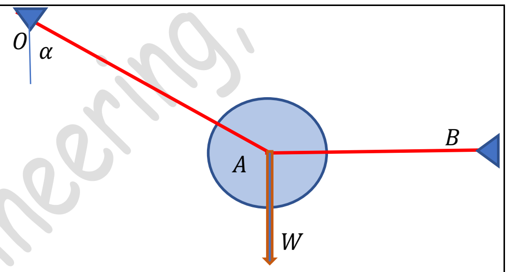
Or

$$\Sigma F_n + \left(-\frac{W}{g}a_n\right) = 0 \quad and \quad \Sigma F_t + \left(-\frac{W}{g}a_t\right) = 0$$

That is, inertia forces of magnitudes equal to the mass times the acceleration is incorporated in to the system in the directions opposite to the directions of accelerations and the body is assumed to be in a state of dynamic equilibrium. Then by DÁlembert's principle, the algebraic sum of external forces and inertia forces in two mutually perpendicular directions are equal to zero.

Example 1

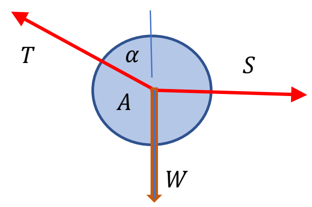
A body of weight W is suspended in a vertical plane by two in-extensible strings as shown. Determine the tension in the inclined string OA an instant before and after the string AB is cut.



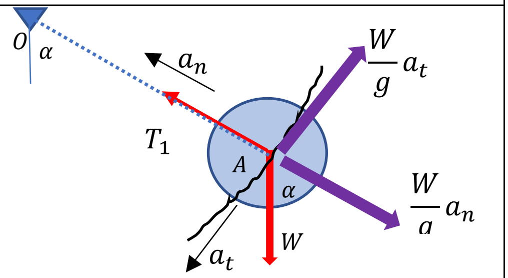
Before the horizontal string AB is cut, the system is in static equilibrium. Let us draw the free-body diagram of the ball.

$$\Sigma F_y = 0 \lll\ggg T\cos\alpha - W = 0 \ll\gg T = W\sec\alpha$$

$$\Sigma F_x = 0 \lll\ggg S - T\sin\alpha = 0 \ll\gg S = W\tan\alpha$$



When the horizontal string AB is cut, the ball starts swinging down in a circular path. Since it is moving along a circular path, the acceleration of the body can be resolved in to normal and tangential components. Inertia forces of magnitude $\frac{W}{g}a_n$ and $\frac{W}{g}a_t$ are incorporated in the free-body diagram in the opposite directions of respective accelerations and the system is in dynamic equilibrium



By DÁlembert's principle, algebraic sum of all the forces (both external and inertia)along two mutually perpendicular directions are zero.

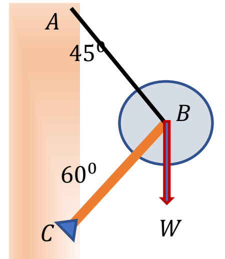
$T_1 - \mathrm{W}\cos\alpha - \frac{W}{g}a_n = 0$ $\frac{W}{g}a_t - W\sin\alpha = 0$	$a_n = \frac{v^2}{\rho} \quad and\ a_t = \frac{dv}{dt}$ But an instant immediately after the string AB is cut, the body starts swinging from down from rest. Therefore, the velocity of the body is zero. Hence, $a_n = 0$

$$T_1 - \mathrm{W}\cos\alpha = 0$$

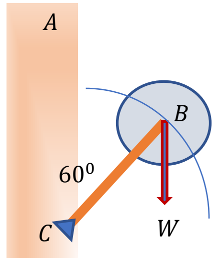
$$T_1 = \mathrm{W}\cos\alpha$$

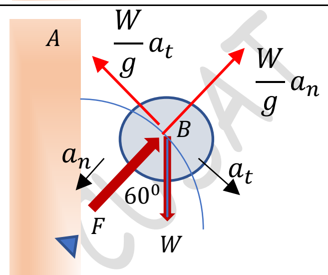
Example 2

A ball of weight W is supported in a vertical plane as shown. Find the compressive force in the bar BC just after the string AB is cut. Neglect the weight of the bar.



When the string AB is cut, the ball starts to swing down in a circular path. The acceleration of the ball can be resolved in to two components a_n and a_t. The normal acceleration is directed towards the centre of rotation and tangential acceleration is directed tangential to the curve in the direction of velocity.





Let us draw the free-body diagram of the ball. Inertia forces of magnitudes $\frac{W}{g}a_n$ and $\frac{W}{g}a_t$ are incorporated in the free-body diagram in the opposite directions of respective accelerations and the system is in dynamic equilibrium. Then, by DÁlembert's principle, algebraic sum of all the forces in the normal and tangential directions are equal to zero.

$$F - W\cos 60 + \frac{W}{g}a_n = 0$$

$$\frac{W}{g}a_t - W\sin 60 = 0$$

$$a_n = \frac{v^2}{\rho} \quad and\ a_t = \frac{dv}{dt}$$

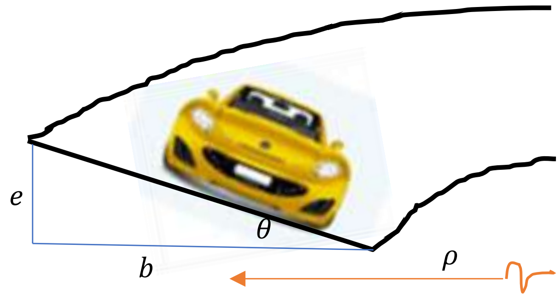
But an instant immediately after the string AB is cut, the body starts swinging from down from rest. Therefore, the velocity of the body is zero. Hence, $a_n = 0$

$$F - W\cos 60 = 0$$

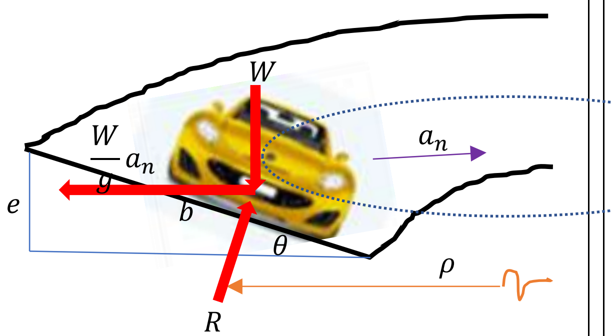
$$F = W\cos 60 = 0.5W$$

Example 3

Determine the banking angle θ of the circular track so that the wheels of the sports car will not have to depend on friction to prevent the car from sliding sideways. The car travels at a constant speed v=30 m/s. The radius of the track is 180m. Neglect the size of the car.



As the car travels along the track, the centre of gravity of the car moves along a horizontal circular path. Hence the normal acceleration is directed towards the centre of the horizontal circle and the normal inertia force is directed opposite to it. Since the speed is uniform, the tangential acceleration is zero. Also as per the question, friction is not considered. Let us draw the free-body diagram. The forces are W, normal reaction R, and the normal inertia force $\frac{W}{g}a_n$



Resolving the forces parallel to the track

$$\frac{W}{g}a_n\cos\theta - W\sin\theta = 0$$

$$\tan\theta = \frac{v^2}{\rho g} = \frac{30^2}{180\times 9.81} = 0.51 \quad \lll\ggg \quad \theta = \tan^{-1}(0.51) = 27^0$$

If b is the width of the track, then the super elevation e can be found out from the relation

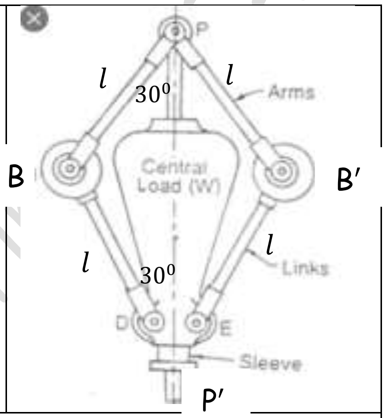
$$\sin\theta = \frac{e}{b}$$

For small values of θ, $\sin\theta \cong \tan\theta$

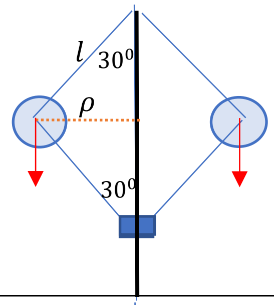
$$e = \frac{bv^2}{\rho g}$$

Example 4

At what uniform speed of rotation around the vertical axis PP′ will the balls B and B′ of equal weight $Q = 45\ N$ begin to lift the weight $W = 90\ N$. Neglect the friction and weights of the hinged bars of length $l = 25\ cm$. The sleeve can slide freely along the shaft. Take the angles the arms make with the vertical shaft as 30^0.

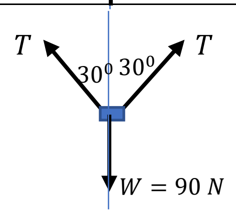


Since the arms are pivoted to the shaft PP′, as the shaft rotates, the balls B and B′ will also move along a horizontal circular path of radius $\rho = l\sin 30 = 0.125\ m$. As the shaft is rotating with uniform speed, the tangential acceleration of the balls will be zero. There will be only normal acceleration component $a_n = \frac{v^2}{\rho}$ directed towards the centre of the circle.

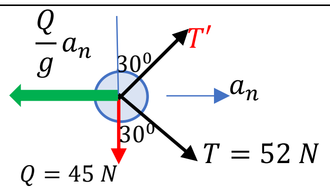


Let us consider the free-body diagram of the sleeve with weight. Resolving the forces in the vertical direction, we write,

$$T\cos 30 + T\cos 30 - W = 0 \lll\ggg \qquad T = \frac{W}{2\cos 30} = 52\ N$$



Now consider the free-body diagram of the ball. Normal inertia force of magnitude $\frac{Q}{g}a_n$ is incorporated in the free-body diagram in the opposite direction of normal acceleration and the ball is in dynamic equilibrium. Then, by DÁlembert's principle, algebraic sum of all the forces in two mutually perpendicular directions are equal to zero



$T\sin 30 + T'\sin 30 - \frac{Q}{g}a_n = 0$ $T' - T\cos 30 - Q = 0$	$T' = \frac{45 + 52\cos 30}{\cos 30} = 104\ N$ $a_n = 17\frac{m}{s^2}$	$\frac{v^2}{\rho} = 17\frac{m}{s^2}$ $v = \sqrt{15.5\times 0.125} = 1.46\ m/s$

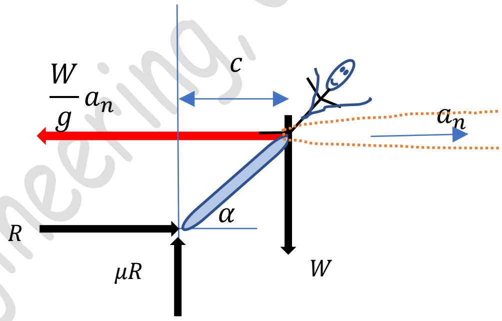
The tangential velocity of the ball moving along a circular path of radius $\rho = 0.125\ m$ is $v = 1.46\frac{m}{s} = \frac{2\pi N}{60}\rho$.

$$\therefore N = 111\ rpm$$

Example 5

A stunt performer rides his motor cycle around the inside of a vertical circular drum of radius r. What is the minimum speed he should have to prevent slipping down if the coefficient of friction between the all and the tyres is μ. When he is riding at this speed, what angle he should make with horizontal in order to prevent tipping down? Take the total weight of the vehicle and rider as W and the combined centre of gravity is at a horizontal distance c from the wall.

Let the rider riding the bike at a uniform velocity v which is just sufficient to prevent him from sliding down. He is leaning at an angle α with the horizontal in order to prevent overturning. As the rider moves in a circular path, the combined centre of gravity moves along a horizontal circular path of radius $(r-c)$. The normal acceleration $a_n = \frac{v^2}{r-c}$ is directed towards the centre of this circle and the tangential acceleration is zero.

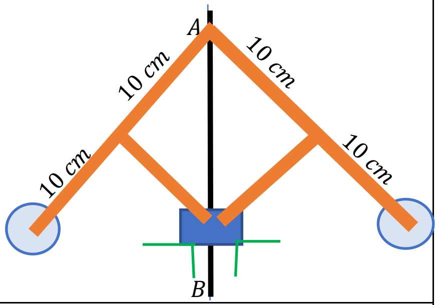


Let us draw the free-body diagram of the system incorporating the normal inertia force $\frac{W}{g}a_n$ in a direction opposite to the direction of the normal acceleration. The other forces are weight W, normal reaction R, and frictional force μR. By DÁlembert's principle, once the inertia forces are incorporated, the system can be solved like a static system. Hence, we can write,

$R - \frac{W}{g}a_n = 0$ $\mu R - W = 0$ $\frac{W}{g}a_n \times c\tan\alpha - W \times c = 0$	$R = \frac{W}{\mu}$ $a_n = \frac{Rg}{W} = \frac{g}{\mu}$ $v = \sqrt{[g(r-c)/\mu]}$	$\tan\alpha = \frac{W \times c \times g}{c \times W \times a_n} = \mu$ $\alpha = \tan^{-1}\mu$

#Exercise 1

The $10\,N$ weight of the governor can slide freely on the vertical shaft AB. At what uniform speed about the axis AB will the $5\,N$ fly balls lift the sliding weight free of its support? Neglect friction and weights of the bars.



#Exercise 2

Racing cars travel around a circular track of radius 300 m with a suniform of 350 km/hr. What angle α should the floor of the track make with the horizontal in order to safegaurd against skidding?